

Nirupathunga M

Coimbatore, Tamilnadu, India.

Phone: +91 93453 20689 Email: nirupathunga@gmail.com (or) nirupathunga.71382402102@sritcbe.ac.in

LinkedIn: www.linkedin.com/in/nirupathunga-m-46b744322

Career Objective

A highly motivated and technically skilled Computer Science student with hands-on experience in Machine Learning (Computer Vision), Autonomous Systems, and Full-stack Development. Demonstrated leadership through securing ₹15 Lakhs in MSME funding for innovative industrial solutions and qualifying as a Semi-Finalist in the Caterpillar Autonomy Challenge at IIT Madras. Seeking an entry-level role to apply advanced problem-solving, AI architecture, and analytical skills to drive technological innovation and develop impactful, real-world solutions.

Education

Qualification	Institution	Details
B.E. Computer Science and Engineering	Sri Ramakrishna Institute of Technology	Expected Graduation: 2028 CGPA: 8.9 (1 st - 3 rd Semester)
12 th Boards CBSE	Kovai Public School	91.2%
10 th Boards CBSE	Kovai Public School	96%

Skills

- Programming Languages: C (IIT Bombay Certified), C++, Java, Python, Object-Oriented Programming (OOP), Data Structures & Algorithms
- Web Development: HTML5, CSS3, JavaScript, Tailwind CSS, Responsive Web Design, REST APIs (Basic)
- Machine Learning & AI: YOLO (v8/v11), Data Preprocessing, Model Training & Evaluation, Scikit-learn, Pandas, Deep Learning (MathWorks Certified), Prompt Engineering
- Software Tools & Platforms: MATLAB & Simulink, Git & GitHub, Adobe Creative Cloud (Photoshop, Illustrator, Premiere Pro), Microsoft Office Suite, Tally ERP
- Embedded Systems & Robotics: Arduino, ESP Microcontrollers, ROS (Basic), SLAM & Navigation, Sensor Integration, IoT System Development
- Soft Skills: Problem Solving, Project Management, Event Coordination (ASME PR Head), Attention to Detail, Team Collaboration, Effective Communication

Projects

- **SmartSync 4.0: AI-Powered Digital Twin Platform**

Apr 2026 – Present

Designed an Industry 4.0 Digital Twin platform for manufacturing innovation, securing **₹15 Lakhs** in government funding through the **MSME Idea Hackathon 5.0**. Integrated real-time data synchronization and AI-driven predictive modeling to optimize manufacturing efficiency for MSME units.

- **YOLO-based AR Guidance System (TN-IMPACT 2026)**

Feb 2026 - Mar 2026

Engineered a real-time object detection system using **YOLO v11** to provide augmented reality guidance for industrial valve assembly. Achieved high-precision detection in variable lighting conditions, reducing assembly error rates during the **TANCAM Industrial Hackathon**.

- **NASA International Space Apps Challenge (Lunar Crater Analysis)**

Oct 2025 - Dec 2025

Built a deep learning pipeline for automated detection and classification of lunar craters using satellite imagery. Recognized as a Galactic Problem Solver for innovative application of computer vision in space exploration and mapping.

- **Autonomous Rover Navigation (Caterpillar Autonomy Challenge)**

Jan 2026 - Mar 2026

Developed autonomous navigation solutions using **SLAM** and **Monocular Vision** for the Caterpillar Autonomy Challenge at **Shaastra, IIT Madras**. Qualified as a **Semi-Finalist** by optimizing computer vision algorithms for obstacle detection and path planning in dynamic environments.

- **AI-Powered Laser QR Code Marking & Monitoring System for Indian Railways Track Fittings**

Aug 2025 - Present | Smart India Hackathon 2025

Developing a portable laser engraving system to mark durable QR/Datamatrix codes on railway track fittings and Developing AI-based defect detection (OpenCV, TensorFlow) to verify engraving quality and flag anomalies. Built a web app for railway staff to scan codes, fetch lifecycle details, and sync with UDM & TMS platforms which delivers analytics dashboards for predictive insights, inspection trends, and warranty management.

- **Facial Emotion Recognition using VGG16 & VGG19**

Oct 2024 - Mar 2025

Built and trained deep learning models for emotion classification using VGG16 and VGG19. Achieved 92% accuracy using transfer learning, data preprocessing, and optimization techniques.

Patents

- **Augmented Reality Based Intelligent Assembly Guidance and Torque Monitoring System**

Status: Patent Filed

Description: Developed an integrated AR system that utilizes **YOLO v11** for real-time component detection and assembly sequence verification. Designed the system to provide visual assembly overlays and monitor torque values, ensuring precision and reducing human error in industrial manufacturing.

Roles and Responsibilities

- **Public Relations Head – ASME SRIT Student Section Feb 2026 – Present**

Leading the strategic communication and outreach initiatives for the ASME student chapter. Managing professional storytelling on LinkedIn and other social platforms to announce technical milestones, coordinate with industry mentors, and enhance the section's global presence within the engineering community.

Skills: Professional Communication, Brand Storytelling, Stakeholder Management, Social Media Strategy.

- **Innovation Ambassador – Hub for Innovation and Entrepreneurship (HIVE) Dec 2024 – Present**

Promoting a culture of creativity and entrepreneurship within the college, encouraging students to explore innovative solutions and startup ideas. Facilitating workshops, collaborating with mentors, and bridging the gap between aspiring innovators and industry resources.

Skills: Innovation Promotion, Entrepreneurship Support, Workshop Facilitation, Startup Ecosystem Engagement.

- **Executive Member - Technical Association of Computer Engineers (TEKACE)**
Jul 2025- Present

Awards & achievements

- **Winner – MSME Idea Hackathon 5.0 (Ministry of MSME, Govt. of India) | 2026**
Secured **₹15 Lakhs** in government funding for the innovation project "**SmartSync 4.0**".
- **Semi-Finalist – Caterpillar Autonomy Challenge (Shaastra, IIT Madras) | 2026**
Recognized for developing autonomous navigation and SLAM algorithms for a robotic rover.

- **All India Rank (AIR) 41 – Aryabhatta National Maths Competition | 2025**
Scored **96.67%** in the national-level analytical and problem-solving examination.
- **Technical Winner – ASME Oral Competition (EFx SRIT) | 2026**
Awarded for presenting research on AR-based Intelligent Assembly systems.
- **Galactic Problem Solver – NASA International Space Apps Challenge | 2025**
Recognized for innovative deep learning applications in lunar surface analysis.

Technical certifications

1. Programming & Core Engineering

C Training – Spoken Tutorial, **IIT Bombay** (2025)
Python 3.4.3 – Spoken Tutorial, **IIT Bombay** (2025)
Linguaskills Business – University of Cambridge (2025)
Cybersecurity Foundation and Operations- LearnLike and CyberVault(2026)

2. AI, Machine Learning & Robotics

MATLAB & Simulink Onramps – MathWorks (2025)
Fundamentals of Encryption & Quantum-Safe Techniques – IBM(2026)
SQL Basics(Standard)- SkillRack(2026)
Curve Fitting Onramp – MathWorks(2026)
Optimization Onramp – MathWorks(2026)

3. Design & Professional Skills

Design Thinking: A Primer – NPTEL, **IIT Madras** (2025)
Introduction to Internet Of Things- NPTEL(2026)
Intellectual Property UTSAV – AICTE(2025)

Mini Projects

- **Fake News Detector | Machine Learning Mar 2025**
Developed a supervised learning model to classify and detect misinformation in news articles using Natural Language Processing (NLP).
Implemented text vectorization and classification algorithms to achieve high accuracy in distinguishing between credible and fabricated content.
- **ZenSpace | Physics-Engine Stress Relief Web App Dec 2025**
Engineered a high-performance web application featuring a custom physics engine and particle system designed for interactive user engagement and stress relief.
Focused on smooth UI/UX rendering and real-time physics calculations for a seamless browser-based experience.

Additional Information

Date of Birth: 15 May, 2006

Languages known: English (Proficient - Cambridge BEC B2 Overall), Tamil (Proficient User)